TARI29 Artificial Intelligence, 7.5 credits. Exam Instructions

Allowed aids:

- VS Code and SWI-Prolog installed at the computers in the exam room
- SWI-Prolog documentation provided as PDF
- The logical equivalence sheet provided as PDF
- Other material provided as PDF
- Calculator

However, you are not allowed to use any other form of assistance. Specifically, it is of course *strictly prohibited* to collaborate with other students or use online resources during the exam.

Instructions for answering questions:

- Each question will provide a task description and the maximum amount of points that can be awarded for a correct answer.
- Type in your answers directly in the Inspera system.
- If a question *mentions* that you can submit a drawing or other type of writing on paper, you can get paper sheets from the personnel in the exam room. Then you need to hand in your sheets to the personnel. The paper sheets will be delivered to the examiners after the exam. If you write on paper, **write in legible (clear) handwriting** so that your answer is well readable.
- For *Prolog problems*, write and debug your code in the VS Code environment. Then copy the Prolog code of your solution from VS Code and paste it directly in the text field of the corresponding question.

Grading:

3	28–41.5 points
4	42–55.5 points
5	56–70 points

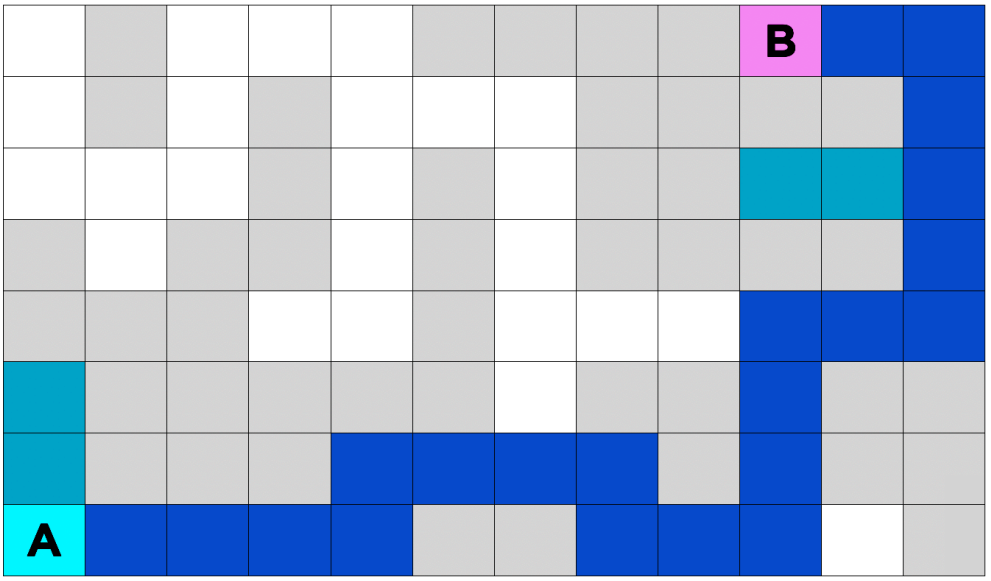
The distribution of points among question sections:

Section	Search	Game Theory	MDP	Logic	Prolog
Points	12	12	9	14	23

Good Luck!

1 Search Algorithms

Consider the maze shown below. Grey cells represent walls. A search algorithm was executed on this maze and produced the dark blue path from point A to point B. The cyan-highlighted cells represent states that were explored during the search but did not lead to the goal.



Four search algorithms were discussed in the lecture. For each of the algorithms listed below, indicate whether it could have produced the observed search behavior (Yes/No), and provide a brief justification for your answer. Be as specific as possible in referring to the properties of the algorithm and the observed search pattern.

1. Depth-First Search
2. Breadth-First Search
3. Greedy Best-First Search with Manhattan distance heuristic
4. A* Search with Manhattan distance heuristic

Fill in your answer here. If you answered on paper, type in here "Answered on paper"

Maximum marks: 12

2 Game Theory - Task 1

1. Consider the payoff matrix below for a game in normal form.

		Player B		
		X	Y	Z
Player A	X	3,4	5,3	2,3
	Y	2,5	3,9	4,6
	Z	3,1	2,5	7,4

   Are there any **pure Nash Equilibria** in the game? If so, what are they (and why)?
2. Below, you see the payoff matrix of the Choosing Paths game, where two drivers meet each other on a road, and must choose to turn left or right.

		Driver 2	
		left	right
Driver 1	left	1,1	0,0
	right	0,0	1,1

   Are there any **Pareto Optimal outcomes** in the game? If so, what are they (and why)?

You may use paper sheets to answer this question.

Fill in your answer here. If you answered on paper, type in here "Answered on paper"

Maximum marks: 5

3 Game Theory - Task 2

Let us assume the following scenario. Two pharmaceutical laboratories, NordPharm and SwedLab, want to develop a new drug. They can either team up and aim to develop a new generation drug, or each can work on its own to develop a less ambitious drug. Based on their decision, the following cases may occur:

- if they team up and agree to develop the new generation drug together, then a better product will be released on the market, leading to profits of 2 billion SEK for each company
- if they defect and work on their own project, they will develop a less ambitious drug, leading to lower profits for both, equal to 1 billion SEK
- but if one cooperates (i.e., starts working on an agreed common project) and the other defects (i.e., even though they have agreed on a common project, this company starts working on their own), then the common project will fail (i.e., no profits) while the latter develops a less ambitious drug and earns a profit of 1 billion SEK.

Represent the above situation as a game in its Normal Form. What is the dominant strategy of each company and why? Can we predict what would be the outcome of this game? Do we make any assumptions to make this prediction?

You may use paper sheets to answer this question.

Fill in your answer here. If you answered on paper, type in here "Answered on paper"

Maximum marks: 3

4 Game Theory - Task 3

Calculate a Nash Equilibrium with mixed strategies from the payoff matrix below and state the value of the game at this equilibrium. Use the method of assuming the opponent should be indifferent between its actions. Show all your calculations.

		Player B	
		X	Y
Player A	X	1,1	3,0
	Y	4,1	2,2

You may use paper sheets to answer this question.

Fill in your answer here. If you answered on paper, type in here "Answered on paper"

Maximum marks: 4

5 MDP / Reinforcement Learning - Task 1

Assume an MDP consists of the two states S_1 and S_2, the two actions A_1 and A_2, a *discount factor* $\gamma = 1/3$, a *reward function* R and a *transition function* T as shown below:

$$R(s,a,s') = \begin{cases} 1 & \text{if } s' = S_1 \\ -1 & \text{if } s' = S_2 \end{cases}$$

s	a	s'	T(s,a,s')
S_1	A_1	S_1	$1/3$
S_1	A_1	S_2	$2/3$
S_1	A_2	S_1	$1/2$
S_1	A_2	S_2	$1/2$
S_2	A_1	S_1	$1/4$
S_2	A_1	S_2	$3/4$
S_2	A_2	S_1	$1/2$
S_2	A_2	S_2	$1/2$

Perform one iteration of <u>value iteration</u> and enter the <u>value</u> for $V_1(S_1)$ in the table below (you don't have to enter a value into the gray cell). Use the initial values for $V_0(s)$ below instead of initializing all values to 0. Show all your calculations.

s	$V_0(s)$	$V_1(s)$
S_1	3	?
S_2	6	

$$V_{k+1}(s) \leftarrow \max_a \sum_{s'} T(s,a,s')[R(s,a,s') + \gamma V_k(s')]$$

You may use paper sheets to answer this question.

Fill in your answer here. If you answered on paper, type in here "Answered on paper"

Maximum marks: 3

6 Understanding Q-learning Parameters

Q-learning involves several parameters that influence how the agent learns:
• *learning rate* (α)
• *discount factor* (γ)
• *exploration rate* (ε in ε-greedy policy)

For each of the three parameters above, explain:
1. What the parameter controls in the learning process
2. How setting it too high or too low might affect learning

Write a few sentences per parameter. Be concise but specific.

Fill in your answer here. If you answered on paper, type in here "Answered on paper"

Maximum marks: 6

7 Logic: First-order logic—Task 1

Consider the following knowledge base, given in natural language:

```
Harry is not evil.
Every wizard knows Harry.
There is an evil wizard who smokes.
Every wizard who knows Harry also knows Dumbledore.
```

Translate the knowledge base into first-order logic, using one FOL sentence for each sentence above.

You must use the following predicates and constants:

- *Unary predicates*; i.e. taking one argument: wizard, smokes, evil.
- *Binary predicates*; i.e. taking two arguments: knows.
- *Constants*; i.e. names of people: Harry, Dumbledore

Logical_equivalences

You may use paper sheets to answer this question.

Alternatively, you can use the following way to quickly type in logical formulae in Inspera.

- For logical connectives, use symbols that are used in the programming languages Java, C\#, and C++: '!' for negation, '&&' for conjunction, '||' for disjunction, '->' for implication , " <--> for biconditional
- For universal quantifier use 'forall', and for existential quantifier use 'exists'.

Fill in your answer here. If you answered on paper, type in here "Answered on paper"

Maximum marks: 4

8 Logic: First-order logic—Task 2

Consider the following knowledge base consisting of a single first-order logic sentence:

$$\forall x\, y.\ \mathrm{knows}(x, y)$$

Define <u>two first-order models</u>:
• One in which this sentence is **true**
• One in which this sentence is **false**

For each model, formally specify:
• The domain
• The interpretation function for the knows predicate

Fill in your answer here. If you answered on paper, type in here "Answered on paper"

Maximum marks: 4

9 Logic: Resolution—Task3

Consider the following knowledge base (KB), given in propositional logic and already in conjunctive normal form (CNF):

1. $A \vee B$
2. $\neg A \vee C$
3. $\neg B \vee C$
4. $\neg C \vee D$
5. $\neg D$

We want to show that the query C is entailed by the knowledge base.
Use resolution to prove that $\mathrm{KB} \models C$. Number your steps and show clearly which clauses are resolved.

Fill in your answer here. If you answered on paper, type in here "Answered on paper"

Maximum marks: 6

10 Prolog: Constraint Logic Programming (CLP)

Write a Prolog program that solves the cryptarithmetic puzzle below using CLP(FD). You need to find all concrete solutions.

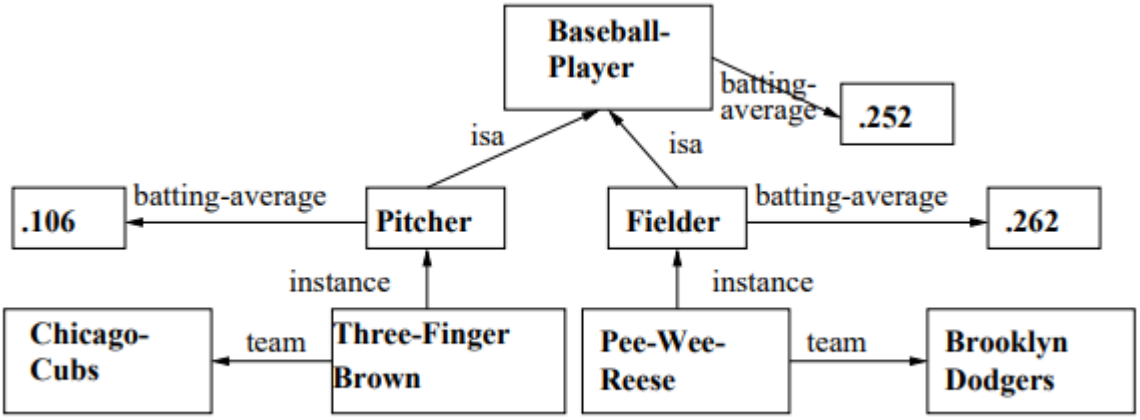
```
  F A N
+ F A N
+ S O F A
----------
H O U S E
```

Copy the Prolog code of your solution from VS Code and paste it directly in this text field.

Maximum marks: 8

11 Prolog: Semantic Networks

In this task you will use semantic network to encode several facts about baseball. The following diagram shows an example of a simple semantic network (source: https://www.professional-ai.com/difference-between-semantic-net-and-frame.html).



The task is to represent this semantic network with Prolog clauses.

Hint: Each triple ‘box–arrow–box’ from the diagram will become a Prolog clause with a binary predicate corresponding to the label of the arrow between the boxes. The boxes themselves will become the arguments.

Copy the Prolog code of your solution from VS Code and paste it directly in this text field.

Maximum marks: 8

12 Prolog: Arithmetic operators

Write a procedure **cubes_until(X, Y)** that simulates a *do-until* loop. The predicate shall write all cubes of numbers starting with *X* and increasing it by one until $X^3 > Y$.

You must use **cut** (!) to exclude futile backtracking.

Copy the Prolog code of your solution from VS Code and paste it directly in this text field.

Maximum marks: 7